

Weekly Goals

- Clean up weather processing pipeline
 - Remove unneeded files and functions
 - Finish documentation
 - Push to main
- Integrate weather api with main model call
- Update SSoT with documentation on weather pipeline
- Test deployed site for any errors
- Begin working on final document

These are the concrete tasks that I want to get done this week. If time permits (or if my team needs) I will add or reassign myself to more functional requirements.

This document will be short. Given exams that are coming up which I need to prepare for, that will cut my time down this week and given that half of my tasks are 'write in a different document,' this report will likely be much shorter than the previous documents. I will do my best to document general tasks I did and what sections of the files I worked on, but it is redundant to be typing into this document about how I wrote in other documents.

Weather Pipeline Cleanup

The easiest thing to do is to mark files for deletion. The following files were created over the course of the last 10 weeks:

- **CreateTrainingData.py:** The 'master file' or the file that needs to be called in order to make the data.
- **csv_overview.py:** A file that I used to get general information on the contents of the crash data. No purpose to data generation pipeline.
- **csv_editing.py:** The pipeline that does the processing of weather and crash data. This is the file that makes the training data when it finishes.
- **csv_lookup.py:** Combs the raw crash data for cities and makes a list of them. The first step in the process of getting weather from multiple sources (the file that chooses where we look for weather)
- **csv_geocode.py:** Took in the isd-history while I was using NOAA data and used latitude and longitude to tell me what city and county each station was in
- **csv_weather.py:** Preprocessing step for the weather when I was using NOAA data. Not updated after the change to Open-Meteo.
- **open_download.py:** This is fed the year and the list of cities and gets weather for the city/coordinates at a specific year. Downloads and renames a csv file to the project
- **open_preprocessing.py:** The newer preprocessing file for the Open-Meteo data. Transforms all the field inside of weather_csvs
- **weather_download.py:** Downloaded weather using the csv_geocode data when NOAA was still being used
- **road_geocaching.py:** Unlike what the file is named, it does not geocache. It gets the road sections lined up and then returns a file with the approximate middle picked out

From these files: csv_geocode, csv_weather, and weather_download can immediately be deleted. They have no functionality inside of CreateTrainingData and thus there is no more reason to have them inside of the directory. Similarly geocoded.csv and isd-history.csv can be removed. These were saved under the assumption that they contained data that would be a hassle to generate and would be used any time new weather needed to be downloaded. This is not the case anymore.

csv_overview can also be deleted. While it provided good insights into the code, none of it was critical to the pipeline being run. Should diving into the data to understand it better be needed again in the future, a more specialized version can be made.

The second thing to do is to go through and ensure that all the functions work and notate what they do in code. Most are well documented. csv_lookup.py had many redundant functions. Most of its code was able to be reused during the transition to Open-Meteo but at least half has been scrapped with its documentation updated.

Limits and things to note

The biggest issue that I think still exists with this data is that I have not worked with it for multiple years yet. Despite there being an array for multiple years to be inputted I have not tested (nor do I think I can accurately test) multiple years of data at present. If I am to finish everything else early, I will attempt to get this feature implemented.

The other limit right now is that a while ago we got a large batch of NOAA data that needs a lot of preprocessing before it could be implemented as training data. This could also be used as our proof of success if the AI could generate probabilities that somewhat match training data for a year it has not been trained on yet. At this point I know that I have no hope of completing this function before the end of the project, even if I were to ignore all of my other objects for the next two weeks. This is not a good use of my time so unless nothing better to do exists, I will ignore it for now.

Final Note

The README and documentation are updated, I will push my branch to main where I worked on everything. There is a possibility that before the project ends I go back and make a new branch, but for now that is not my intent.

Weather API

The other bit of coding I directly planned to do this week (assume no bugs in the application) was to combine the weather API I had written a few weeks ago with the model call. That way the model can fetch current weather quickly before it needs to run.

Upon looking through the code, someone else on the team has written a weather service that is now acting in the place of my weather API using weather.gov and Open-Meteo as sources. The format of how the information is given to the model is vastly different from the way that my weather api is modeled after.

There is no apparent caching or any type of scalability with this new weather service as far as I can tell, but the group agrees that it is good enough and with limited time remaining we have agreed that no major changes should be made, especially if we don't think we can get them tested and working in time.

Given that this service exists, I have been relieved on working on this I suppose. From here on out, I will be mainly working on documentation and preparing for the final presentation.

General Document Writing

There is no point delving deep into what I write in other documents here, but I will say that I worked on:

- SSoT: Updating AI data structures and past requests such that it would actually be able to reproduce (to some degree) what I need it to.
- Project Report: Filling in areas that have no text and editing some areas that do. Right now I am mainly focused on ensuring that everything that I can fill out is filled out. There are some areas that I can contribute to but will need to inform everyone else about so they can help fill things in.

These will be done while testing is done. For this week I want to fill out as much as I can inside of both documents and then can work on editing and rewrites to make things so more cohesive with the rest of my team next week.

Testing

One of the tasks that has been assigned to generally everyone is to test the application to ensure that there are no last minute bugs or unexpected issues ahead of the demo.

Going through the deployed version of the code has given me a couple of different things to maybe improve on:

- Some boxes in the report section stay bright white in dark mode making it hard to read, these should be fixed

- When inputting a road name, sometimes the dropdown just vanishes, hard to know if the road exists or not. It would be good to have some sort of loading indicator.
- The road report screen doesn't refresh very well. It keeps the state of the last request while it fetches the new state. Not critical but worth looking into.
- Runtime note inside methodology is stuck on dark mode
- I don't know if it is intentional or not but the github, TxDot, and Weather.gov in the footer do not actually link anywhere/are plain text.
- On the homepage, the "Get ready for a safer journey" box is stuck in dark mode (I assume) and has an interactable button that has no function.
- The box containing the search bar on the main page is stuck in light mode even when in dark mode
- Not an error, but the search bar could really use a clear feature.

At present these are minor front-end improvements or bugs. General bug searching is also something that everyone on the team has been tasked with looking for. Regardless, I will make these alterations or at least see if some of these alterations have already been made by the time I get to them.

I have gone through and fixed all the visual issues. Upon reflection and inspection of the code, it does appear that most of the containers or elements that I found 'stuck in dark mode' did in fact change to some minor degree and thus are likely intentional designs.

Regardless, I have fixed all of what I perceive to be light versus dark mode errors. Loading states and clearing states from the frontend will prove more difficult and thus I will likely save it for imminent work during the next week.

Next Week's Tasks

I do apologize for how short this document is. Most of my time was document work or planning out the documents and thus there was just not as much to write here as I normally do.

- Finish SSoT
 - Need to have a discussion with the team improving the level of detail specification inside the SSoT
 - Add phase 5
 - Verify that everything is inside is complete and cohesive
- Finish Report
 - Ensure that all team members contribute to the overall document and put in specific information
 - Edit and ensure it is all correct and answering the questions it needs to answer
- Create presentation
 - Create slides
 - Create script/notes for the final project
 - Sample Q/A questions
 - Overall just be ready for next Friday
- Add some small frontend changes
 - As time permits but most likely will not happen